

COSC 2030 (CS2) Pre-course experience survey

Time requirement: 10 minutes

Purpose: We use this pre-course experience survey for the purpose of:

- Understanding your preparedness for this course
- Customizing content
- Aiding in engagement and motivation
- Managing class time efficiently
- Implementing differentiation strategies
- Building on prior knowledge

If you have any questions about this survey, please contact Charlotte Gruner, charlotte.gruner@caspercollege.edu

1. Email *

Please rate your experience with using each of the following in your programs.

The questions below ask about your prior experience level. It is OK to answer "No experience at all". You can succeed in this course without having seen any of these topics since many of these topics will be taught during this course.

2. Classes and objects

Mark only one oval.

1 2 3 4 5 6 7

No € ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely experienced

3. Searching algorithms

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

4. Sorting algorithms

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

5. Recursion

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

6. Template classes/generics

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

7. Lists/Linked lists

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

8. Stacks

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

9. Queues

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

10. Priority queues

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

11. Hashing / Hash tables

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

12. Trees

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

13. Binary search trees

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

14. Balanced trees / AVL trees

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

15. Heaps

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

16. Treap

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

17. Graphs

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

18. B-Trees

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

19. Sets

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

20. Associative array, map, or dictionary

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

Please rate your experience with the following

21. Writing test cases

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

22. Algorithm efficiency and Big-O notation

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

23. Choosing the correct data structure

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

24. Writing classes

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

Rate your experience with the following

25. **Problem decomposition:** Breaking the desired functionality into separate parts (e.g. functions) that can be implemented separately.*Mark only one oval.*

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

26. **Parallel processing:** Computing where the operations within a program are being executed simultaneously (i.e. running on different cores of the processor at the same time.)

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

27. **Distributed computing:** Computing with a program that runs using data from a different computer (i.e. an online service)

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

28. **Event handling:** Having a function that responds to an independent action (i.e. a mouse click)

Mark only one oval.

	1	2	3	4	5	6	7	
No €	☐	☐	☐	☐	☐	☐	☐	Extremely experienced

To what extent do you agree or disagree with the following statements

29. In general my interest in computing is high.

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	☐	☐	☐	☐	☐	☐	☐	Strongly agree

30. My interest in learning about parallel and distributed computing is high.

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	☐	☐	☐	☐	☐	☐	☐	Strongly agree

31. In general, I believe learning about parallel and distributed computing is important.

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	☐	☐	☐	☐	☐	☐	☐	Strongly agree

32. I believe I have a good understanding of computing

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	☐	☐	☐	☐	☐	☐	☐	Strongly agree

33. I believe I have a good understanding of parallel and distributed computing.

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	☐	☐	☐	☐	☐	☐	☐	Strongly agree

34. What grade do you expect to get in COSC 2030

Mark only one oval.

- ☐ A (90 - 100)
- ☐ B (80 - 89.9)
- ☐ C (70 - 79.9)
- ☐ D (60 - 69.9)
- ☐ F (0 - 59.9)

35. What specific skills or abilities do you aspire to acquire or improve upon as a result of your learning in this class?

36. When did you first begin attending Casper College?

Mark only one oval.

- ☐ Before 2020
- ☐ 2020-2021 academic year
- ☐ 2021-2022 academic year
- ☐ 2022-2023 academic year
- ☐ 2023-2024 academic year
- ☐ 2024-2025 academic year

37. What is your current class standing?

Mark only one oval.

- ☐ First year (freshman)
- ☐ Sophomore
- ☐ Other:

38. Please indicate your ethnicity (optional)

Mark only one oval.

- ☐ American Indian or Alaska Native
- ☐ Asian/Pacific Islander
- ☐ Black/African American
- ☐ Hispanic
- ☐ Middle Eastern or North African
- ☐ White

39. Please indicate your gender (optional)

Mark only one oval.

- ☐ Male
- ☐ Female
- ☐ Non-binary/ third gender

40. What is your major (choose all that apply)

Check all that apply.

- ☐ Computer Science
- ☐ Engineering
- ☐ Biology
- ☐ Cybersecurity
- ☐ Physics
- ☐ Mathematics
- ☐ Other: _____

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